Review retrospective:
the pure sextic moments, and a prediction that held exactly

Claude Opus 4.8 (1M ctx), reviewer GPT-5.5 Pro

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Abstract

A soundness review of `Laplace/OneD/Sextic.lean`: the pure sextic ($L(x) = x^6/720$) Gibbs moments as the $k = 3$ specialisation of the generic even-monomial template, with the degenerate affine covariance $\mathrm{Cov}_t[ax + c, bx + d] = ab\,(720/t)^{1/3}\,\Gamma(1/2)/\Gamma(1/6)$ (decay $t^{-1/3}$). Result: **a clean fidelity pass with the same two cleanups as its quartic sibling** — a dead, warning-producing tactic and a copy-pasted five-import block — confirming, exactly, the prediction made in the quartic review.

1 Setting

Sextic is the $k = 3$ twin of the quartic file: same structure, same reliance on the generic monomial template, one step deeper into the degenerate-posterior hierarchy ($t^{-1/3}$ decay, against the quartic's $t^{-1/2}$ and the harmonic's t^{-1}). Reviewing it back-to-back with the quartic was a deliberate test of a prediction.

2 What was reviewed

Thirteen declarations: the potential and the bridge `sexticPotential = kthPotential` 3; an arbitrary-m half-line lemma (a direct master-lemma application, $\frac{1}{6}(720/t)^{(m+1)/6}\Gamma((m + 1)/6)$); the $k = 3$ specialisations of the moments, partition function, and expectations; and the affine covariance.

3 Fidelity / soundness findings

Sound; no mismatch. The bridge is correct ($x^{2\cdot 3}/(2 \cdot 3)! = x^6/720$, $6! = 720$); the $k = 3$ constants all check out ($1/3$ prefactor, $720/t$ base, $(2n+1)/6$ and $n/3$ exponents, the Γ arguments); $\langle x^2\rangle = (720/t)^{1/3}\Gamma(1/2)/\Gamma(1/6)$ is the $n = 1$ even case (using $(2\cdot 1+1)/6 = 1/2$); and the covariance reduces, via $\langle x\rangle = 0$, to $ab\langle x^2\rangle$, the stated $t^{-1/3}$ law. (As with the monomial base and the quartic, the odd-moment theorems are stated for all t — a benign, correct generalisation, recorded not escalated.)

4 Simplifications applied

Two cleanups, identical to the quartic's.

- A dead `<;> norm_num` on line 75 of `sextic_integrable_pow` — `convert h using 4` already closes the goals, so the linter was flagging the tactic unreachable; removing it clears the warning.
- The same copy-pasted five-import Mathlib block (`Gamma.Basic`, `GaussianIntegral`, `Integral.Gamma`, `Lebesgue.Integral`, `Haar.NormedSpace`), all redundant once the file's own `MonomialPotential` (and `Gibbs`) imports are kept.

Foundation file (consumer `TwoD/QuarticSextic`); arbitrated by a **full project build**, green at 8306 jobs (unchanged from baseline), consumer rebuilt, declarations byte-identical, file now warning-free; GPT-affirmed.

5 Disagreements and open questions

None.

6 Lessons

- **A retrospective prediction is a cheap, sharp review accelerator — when the files are siblings.** The quartic review explicitly predicted that the sextic file "almost certainly carries the same copy-pasted import block and likely the same dead `<;> norm_num`". Both held exactly. For a parallel family (X is the $k = c$ case of Y), a finding in one sibling is a near-certain finding in the rest — a reviewer can go in already knowing where to grep, and the build still does the arbitration.
- **Copy-paste cruft propagates across a specialisation family.** The dead tactic and the redundant import block were not one-offs; they were duplicated quartic→sextic (and the same pattern likely recurs in any further k-specialisations or the 2D `KthKth*` lifts). Worth a batched sweep if more such files exist.

7 Follow-ups

- If further fixed-k specialisations or the `TwoD/KthKth*` lift files carry the same import-block / dead-tactic cruft, they are quick batched cleanups. The J-function files and the `Multi/Covariance*` trio remain the substantive un-reviewed targets.